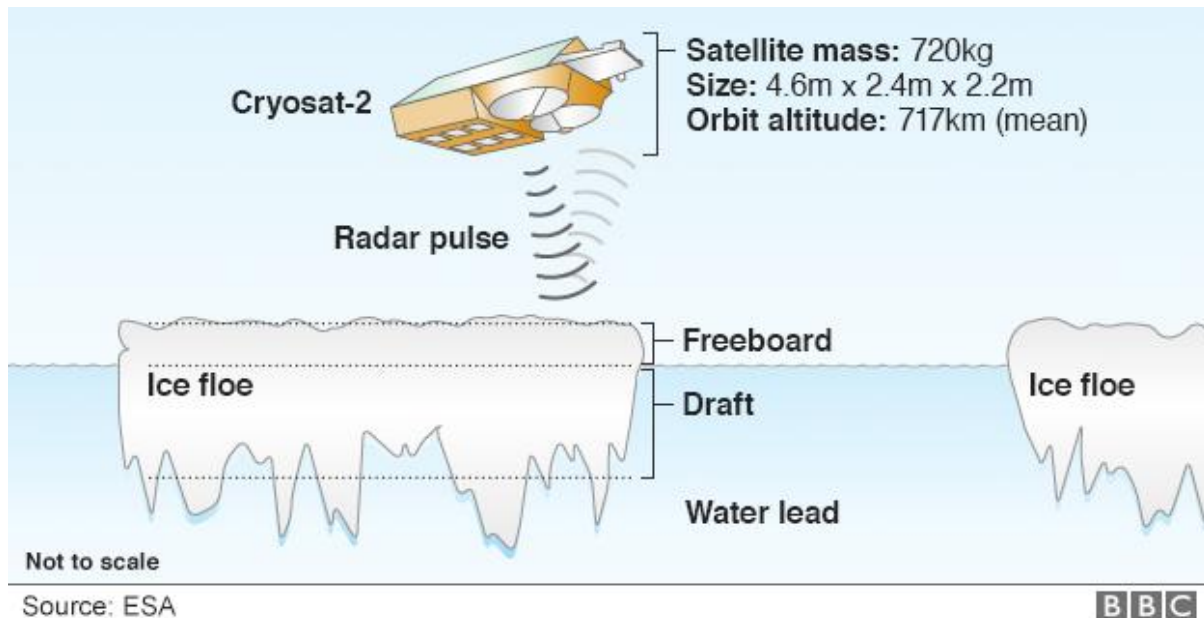


Tutorial-05 Solution

Cryosat-2 problem specification - selection

Given:



The radar senses the height of the freeboard - the ice above the waterline – it is the top $\frac{1}{9}$ th of the floe. The snow and ice on the top of the freeboard is rarely a flat top surface and usually has troughs and peaks.

Knowing this $\frac{1}{9}$ th figure allows Cryosat to work out sea-ice thickness

Some $\frac{8}{9}$ ^{ths} of the ice tends to sit below the waterline - the draft

The thickness multiplied by the area of ice cover produces a volume

Accept three different readings for the freeboard and calculate the average.
(Don't use iteration to capture the readings at this stage)

If the average is below a certain threshold then discard the reading as being statistically insignificant else continue with the calculation of volume.

Output the freeboard, the draft, the area, the thickness, the volume

#No initialisation

1. input

- 1.1. input freeboard-1, freeboard-2, freeboard-3
- 1.2 input length
- 1.3 input breadth

2. process

- 2.1 calc $\text{freeboard_avge} = (\text{freeboard-1} + \text{freeboard-2} + \text{freeboard-3}) / 3$
- 2.2 if $\text{freeboard_avge} < \text{threshold}$ then
 - 2.2.1 discard reading
- 2.3 else
 - 2.3.1 calc $\text{thickness} = \text{freeboard_avge} * 9$
 - 2.3.2 calc $\text{draft} = \text{thickness} - \text{freeboard_avge}$
 - 2.3.3 calc $\text{area} = \text{length} * \text{breadth}$
 - 2.3.4 calc $\text{volume} = \text{thickness} * \text{area}$

3. output

- 3.1 if $\text{freeboard_avge} < \text{threshold}$ then
 - 3.1.1 exit program
- 3.2 else
 - 3.2.1 print freeboard
 - 3.2.2 print draft
 - 3.2.3 print area
 - 3.2.4 print thickness
 - 3.2.5 print volume

Some variation between individual designs is normal, e.g. optimising the calculations. However, this is a reference point for programming so optimisation should not be done to the point of losing clarity, i.e. the calculation etc is too hard or complex to understand.

All variants of the pseudocode should all address the same issue - indicate the flow of logic required to meet the specification and identify key inputs.